

PHY 115L

Woods-Saxon Model for the Nucleus using Variational Methods

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March 12, 2025

1 Introduction

In this assignment we will develop a numerical model for the atomic nucleus, based on the Woods-Saxon potential, an effective potential for the nuclear strong force which binds the nucleus together:

We'll use the Woods-Saxon potential:

$$V(r) = -\frac{V_0}{1 + \exp\left(\frac{r-R}{a}\right)} \quad (1)$$

where:

$$R = r_0 A^{1/3} \quad (2)$$

and A is the mass number. Throughout this exercise, we will be using these values for the parameters:

$$\begin{aligned} V_0 &= 47.78 \text{ MeV} \\ r_0 &= 1.285 \text{ fm} \\ a &= 0.65 \text{ fm} \end{aligned}$$

In principle, the parameters can be tuned differently for protons and neutrons, but we will be using the same values for both nucleons.

An example of this potential is plotted in Fig. 1. It is an effective potential for the nucleus which represents the nucleus as a whole (minus the one single nucleon under consideration). It is appropriate for the nuclear strong force, which is strong but only at short distances.

We will also need to account for the Coloumb repulsion experienced by proton. We will model the charge distribution in the nucleus as a uniform sphere of radius R (with R as above). So this adds an additional term to the potential (for protons only!):

$$V(r) = (Z - 1) \frac{e^2}{4\pi\epsilon_0} \frac{F(r/R)}{R}$$

with dimensionless function:

$$F(x) \equiv \begin{cases} \frac{3-x^2}{2} & r \leq 1 \\ 1/x & r > 1 \end{cases}$$

which accounts for the fraction of the total charge $(Z - 1)e$ enclosed in a sphere at or radius r .

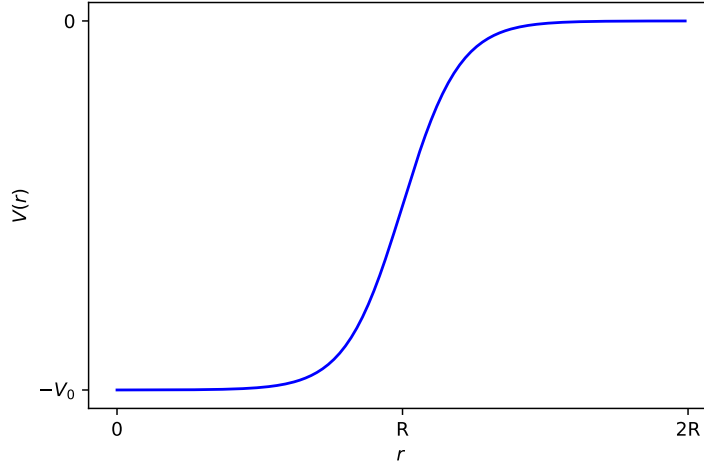


Figure 1: The Woods-Saxon potential for $a/R = 0.1$.

2 System of Units

We will be looking at neutrons and protons (each with a different mass) and using their reduced mass (context dependent). Our system of units from previous assignments includes the mass of the particle in the definition of the energy scale, which is now quite unworkable for us.

We will base our new system of units around two quantities:

$$\hbar c = 197.326 \text{ MeV fm}$$

and the fine structure constant:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{1}{137.035}$$

Atomic nuclei range in size from 1.6 fm to 15 fm. Therefore, we will set our length scale to:

$$L \equiv 1 \text{ fm}$$

We will set our energy scale to:

$$\epsilon = \frac{\hbar c}{L} = 197.326 \text{ MeV}$$

The Hamiltonian operator:

$$\hat{H} = -\frac{\hbar^2}{2M} \frac{d^2}{dx^2} + V(x) \quad (3)$$

becomes:

$$\frac{\hat{H}}{\epsilon} = -\frac{L^2}{2\mu} \frac{d^2}{dx^2} + \frac{V(x)}{\epsilon} \quad (4)$$

where μ is dimensionless quantity related to the mass M by:

$$\mu = \frac{Mc^2}{\epsilon}$$

The masses of the nucleons (neutrons and protons) are:

$$\begin{aligned} m_p &= 938.272 \text{ MeV}/c^2 \\ m_n &= 939.565 \text{ MeV}/c^2 \end{aligned}$$

so in our units:

$$\begin{aligned}\mu_p &= 4.75493 \\ \mu_n &= 4.76148\end{aligned}$$

As before, we will be considering **non-periodic** wave functions. To use the Fourier series approach in this case, we take the period a to be large compared to L , and use the fact that the wave function approaches zero to approximate a periodic function.

The Fourier series is an expansion in plane waves with momentum $p = \hbar k$ and wavenumbers:

$$k = \frac{2\pi m}{a}$$

And so the kinetic energy $K(m)$ of a particle described by the plane wave for coefficient c_m is:

$$K = \frac{p^2}{2M} = \frac{(\hbar k)^2}{2M}$$

and so in our units:

$$\frac{K}{\epsilon} = \frac{1}{2\mu} k^2 L^2 = \left(\frac{1}{2\mu}\right) (2\pi m)^2 \left(\frac{L}{a}\right)^2$$

This differs from our previous exercises by the factor:

$$\left(\frac{1}{2\mu}\right)$$

which you will have to add to your code.

Using the method of separation of variables (e.g. Griffiths Chapter 4) we write the total wave function as:

$$\Psi_{nlm}(r, \theta, \phi) = \frac{u_{nl}(r)}{r} Y_{lm}(\theta, \phi)$$

for quantum numbers n, l, m . The radial part satisfies

$$\frac{\hat{E}}{\epsilon} u(r) = -\frac{L^2}{2\mu} \frac{d^2 u}{dr^2} + \left(V(r) + \frac{1}{2\mu} l(l+1) \frac{L^2}{r^2} \right)$$

Notice that this looks exactly like a 1-D particle, but with an effective potential:

$$\frac{V_{\text{eff}}(r)}{\epsilon} = V(r) + \frac{1}{2\mu} l(l+1) \frac{L^2}{r^2}$$

The term proportional to $l(l+1)$ is called the centrifugal term, and it accounts for the additional kinetic energy in the non-radial direction, due to the angular momentum of the particle. Not surprisingly (since it is due to kinetic energy) it differs from our previous exercise by the same factor

$$\left(\frac{1}{2\mu}\right)$$

which you will have to add to your code as well.

In our system of units, the Woods-Saxon potential becomes:

$$\frac{V(r/L)}{\epsilon} = -\frac{V_0/\epsilon}{1 + \exp\left(\frac{r/L - R/L}{a/L}\right)} \quad (5)$$

where

$$R/L = (r_0/L)A^{1/3}$$

and the parameters were discussed previously.

In our system of units, the Coulomb potential becomes:

$$\frac{V(r/L)}{\epsilon} = (Z - 1) \alpha \left(\frac{L}{R} \right) F(r/R)$$

Note that when reporting energy values to compare with published results (in MeV) you will need to scale by ϵ . (This differs from previous exercises where we stayed exclusively in the dimensionless system of units.)

3 Homework Problems

Please submit one PDF file (including output) and one python notebook file (.ipynb) of your completed assignment. If using C/C++, submit text or PDF of your output and the source code.

Problem 1: Plot the total effective potential (Woods-Saxon, Centrifugal Barrier, and Coulomb potential) for neutrons and protons with $l = 0$ on the same graph. Plot the same quantities on a second graph for $l = 1$. Use the typical parameters from the writeup for both neutrons and protons.

Problem 2: We'll reproduce the results in <https://arxiv.org/pdf/1910.03808> for the 1S and 2S neutron orbitals of ^{56}Fe . Set the parameters to

$$\begin{aligned} V_0/\epsilon &= 47.78\text{MeV}/197.3 \text{ MeV} \\ r_0/L &= 1.285 \\ a/L &= 0.65 \end{aligned}$$

as specified in the paper. For $l = 0$ (no centrifugal barrier) and no Coulomb term, the ground state energy should be -38.3 MeV (this is the 1S orbital). The next state with $l = 0$ should be -18.2 MeV (this is the 2S orbital).

Problem 3: Using the same parameters as in the previous problem (for both neutrons and protons) calculate the energies for neutron and proton orbitals 1S,1P,1D,1F,2S and 2P of ^{56}Fe . Produce an energy level diagram, with neutrons on the left and protons on the right. Draw a line at each energy level and label it with the orbital and occupancy (e.g. "1P — 6").

Problem 4: From your plot, you should be able to discern the magic numbers for nuclei: 2,8,20, and 40. Isotopes with a number of protons or neutrons equal to a magic number tend to be more stable. The magic numbers should correspond to numbers that completely fill energy levels below a large gap. Label each gap in your plot by the corresponding magic number. (Note: we are neglecting spin-orbit coupling which effects things dramatically above the magic number of 20)

Problem 5: Using the energy levels for an ^{56}Fe core with binding energy 492 MeV, estimate the binding energies for ^{57}Fe (500 MeV) and ^{57}Co (498 MeV) by adding the binding energy for a proton

or neutron from the appropriate energy level.

Problem 6: Using the energy levels for an ^{56}Fe core with binding energy 492 MeV, estimate the binding energies for ^{55}Fe (483 MeV) and ^{55}Mn (482 MeV) by adding the binding energy for a proton or neutron from the appropriate energy level.

Problem 7: (Problem dropped: this will not work properly without spin-orbit coupling)

Problem 8: Our simple model with only three parameters does reasonably well on light nuclei as well. Using the energy levels of an ^{16}O core (with binding energy 127 MeV, estimate the energy levels of the isotopes: ^{17}O (131 MeV), ^{17}F (128 MeV), ^{15}O (111 MeV) and ^{15}N (115 MeV).

Problem 9: ^{16}O is a doubly magic nucleus (8 protons and 8 neutrons). Show that it is indeed quite stable by estimating the binding energies of the isotopes: ^{16}F (111 MeV) and ^{16}N (118 MeV).